# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** Assertion (A): The Earth's inner core is solid even though it is extremely hot. Reason (R): The very high pressure at the centre prevents it from melting. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**2.** The core of the Earth has a radius about 3400 km and the Earth about 6400 km. The ratio of the core radius to the Earth radius is about: 1

**(a)** 0.64 **(b)** 0.34 **(c)** 1.88 **(d)** 0.53

**3.** A glacier's melting adds water to the oceans. If all land ice melted, the effect on coastal cities would be: 1

**(a)** drought on the coast
**(b)** flooding by rising sea levels
**(c)** no change
**(d)** cooler climate

**4.** Assertion (A): The temperature in the stratosphere rises with height. Reason (R): The ozone layer absorbs ultraviolet rays from the Sun. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**5.** If the volume of water in oceans is about 1.35 billion cubic km and total fresh water is 3 per cent of total water, with total water about 1.4 billion cubic km, the fresh water volume is about: 1

**(a)** 42 million cubic km
**(b)** 420 thousand cubic km
**(c)** 4.2 billion cubic km
**(d)** 1.4 million cubic km

**6.** Which is the most effective individual action to reduce carbon dioxide in the atmosphere? 1

**(a)** burning waste
**(b)** using public transport and saving electricity
**(c)** using more fuel
**(d)** cutting trees

**7.** Assertion (A): Venus is hotter than Mercury though it is farther from the Sun. Reason (R): Venus has a thick atmosphere rich in carbon dioxide that traps heat. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**8.** A student says the Earth's inner core is liquid because it is hotter than the outer core. This reasoning is wrong because: 1

**(a)** the outer core is cooler
**(b)** pressure raises the melting point
**(c)** iron cannot be solid
**(d)** the inner core is gas

**9.** A town draws more water from wells than rain recharges. In the long term: 1

**(a)** the water table rises **(b)** the water table falls

**(c)** ozone increases **(d)** wells overflow

**10.** Assertion (A): The Moon has large changes in surface temperature between day and night. Reason (R): The Moon has a thick atmosphere. 1

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** The core of the Earth accounts for 15 per cent of the Earth's volume. If the total volume is about 1.08 x 10^12 cubic km find the volume of the core (approx). 2

**12.** Why is the exosphere considered the boundary between the atmosphere and space? 2

**13.** Why are mountain areas colder than the plains even though they are nearer the Sun? 2

**14.** Why does the Earth, though it receives only a fraction of the Sun's energy, have a steady temperature? 2

## SECTION — C (Short Answer Type) 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Describe how Earth's layers formed by differentiation and what evidence supports a liquid outer core. 3

**16.** Explain how the Earth's atmosphere developed over time from the early atmosphere to the oxygen-rich present. 3

**17.** Explain how the Earth's rotation would change the length of day and night if its speed doubled, and its effect on life. 3

OR

Estimate the percentage of the Earth's radius that is made up of the crust given that the thickest continental crust is 70 km and Earth's radius is 6400 km. 3

**18.** Explain how the biosphere depends upon the other three spheres with examples. 3

OR

Explain how soil is formed from rocks and how it supports life. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** The Earth's radius is 6400 km. Using the crust 35 km, mantle 2900 km, outer core 2200 km, find the radius of the inner core, the percentage of radius for each layer and comment on which layer is thickest. 5

OR

Compare the effects of increased CO2 on the oceans (acidification), the atmosphere (warming) and plants (growth). Discuss the benefits and the harms. 5

**20.** Analyse the data: Venus (460 degrees Celsius, thick CO2), Earth (15 degrees Celsius, thin atmosphere), Mars (-60 degrees Celsius, very thin). Explain the role of atmosphere and distance in these temperatures, and what Earth teaches about the greenhouse effect. 5

OR

Explain how scientists use satellites to monitor the Earth: the ozone hole, forest cover, glaciers and cyclones, with two benefits. 5

## END OF QUESTION PAPER Total: 40 Marks